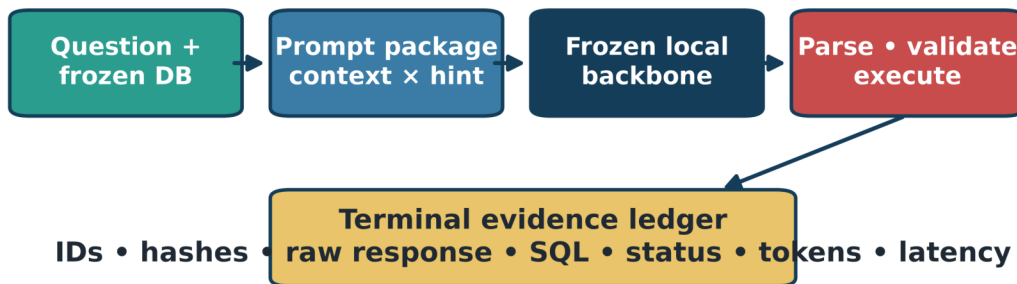


MA-SQLGrid framework figures at approximate MDPI text width

ma_r1_f01_executed_pipeline

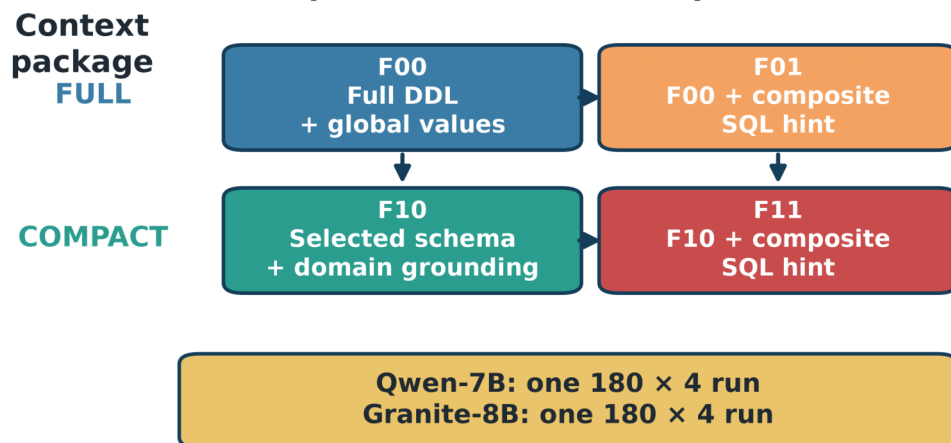
Executed one-pass MA-SQLGrid pipeline



One generation per question/cell/backbone • no candidate ranking • no repair loop

ma_r1_f02_factorial_design

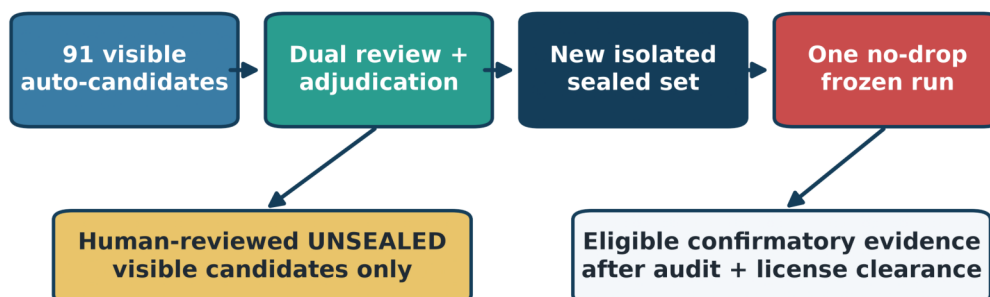
Frozen 2 × 2 prompt-package experiment



1,440 canonical predictions total; “context” is a bundled package intervention

ma_r1_f03_external_evidence_gate

Evidence promotion and sealed external gate



No failed item dropped • separate dataset reporting • access log • permanent artifact